

E-LEARNING MANAGEMENT SYSTEM

DBMS Mini Project - Project Information & PPT Content

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Project Repository :

GitHub : <https://github.com/shidorenikhil56-art/DBMS-Project.git>

Project technologies :

MYSQL/HTML/CSS/JAVASCRIPT/PYTHON OR JAVA

1. Introduction

The E-Learning Management System is a database-oriented application designed to manage online learning activities in an organized and efficient manner. It provides a common platform for students, instructors, courses, lessons, enrollments, assignments, quizzes and results.

The system uses a relational database to store academic information and maintain relationships between users and learning resources. The project demonstrates important DBMS concepts including tables, primary keys, foreign keys, CRUD operations, joins, subqueries, normalization, views, triggers and stored procedures.

2. Problem Statement

Traditional learning management through separate records, spreadsheets or manual processes can make it difficult to maintain student information, course details, enrollment records, assignments and assessment results.

An E-Learning Management System can centralize these records in a relational database. It can provide structured storage, faster retrieval, consistent data and useful academic reports.

- Maintain student and instructor information.
- Manage courses, lessons and learning materials.
- Record student course enrollments.
- Manage assignments, quizzes and submissions.
- Store marks, grades and assessment results.
- Provide reports and search facilities.

3. Objectives

- Create a centralized database for e-learning activities.
- Maintain student, instructor and course information.
- Allow students to enroll in available courses.
- Store lessons and learning resources for each course.
- Manage assignments, quizzes and student submissions.
- Maintain assessment marks and final results.
- Use primary keys and foreign keys to maintain relationships.
- Demonstrate CRUD operations, joins and subqueries.
- Apply normalization up to Third Normal Form (3NF).
- Implement DBMS features such as views, triggers and stored procedures.

4. Database Design

The system is divided into related entities so that each type of information is stored in an appropriate table. This reduces duplication and makes the database easier to maintain.

No.	Table	Purpose
1	Users	Stores student/instructor/admin information
2	Courses	Stores course information
3	Enrollments	Connects students with courses
4	Lessons	Stores lessons belonging to courses
5	Assignments	Stores course assignments
6	Submissions	Stores student assignment submissions
7	Quizzes	Stores quiz information
8	Quiz_Results	Stores student quiz results
9	Payments	Stores course payment information
10	Notifications	Stores notifications for users

5. Tables, Primary Keys and Foreign Keys

Primary keys uniquely identify records in a table. Foreign keys connect records between related tables and help maintain referential integrity.

Table	Primary Key	Important Foreign Keys
Users	user_id	—
Courses	course_id	instructor_id → Users
Enrollments	enrollment_id	student_id → Users, course_id → Courses
Lessons	lesson_id	course_id → Courses
Assignments	assignment_id	course_id → Courses
Submissions	submission_id	assignment_id → Assignments, student_id → Users
Quizzes	quiz_id	course_id → Courses
Quiz_Results	result_id	quiz_id → Quizzes, student_id → Users
Payments	payment_id	student_id → Users, course_id → Courses
Notifications	notification_id	user_id → Users

6. Key Relationships

- Users → Courses: An instructor can create or manage multiple courses.
- Users → Enrollments: A student can enroll in multiple courses.
- Courses → Enrollments: A course can have many enrolled students.
- Courses → Lessons: One course can contain many lessons.
- Courses → Assignments: A course can contain multiple assignments.
- Assignments → Submissions: One assignment can have many student submissions.
- Courses → Quizzes: A course can contain multiple quizzes.
- Quizzes → Quiz_Results: A quiz can have results for many students.
- Users → Notifications: A user can receive multiple notifications.
- Users / Courses → Payments: Payment records connect a student with a course.

7. ER Diagram

The conceptual ER diagram below shows the major entities, attributes and relationships of the E-Learning Management System.

USERS	COURSES	ENROLLMENTS
PK user_id name email role	PK course_id title instructor_id	PK enrollment_id student_id course_id
↓	↙ ↓ ↘	↓
LESSONS	ASSIGNMENTS	SUBMISSIONS
PK lesson_id course_id title	PK assignment_id course_id title	PK submission_id assignment_id student_id
↓	↓	
QUIZZES	QUIZ_RESULTS	PAYMENTS
PK quiz_id course_id title	PK result_id quiz_id student_id marks	PK payment_id student_id course_id amount

PK = Primary Key. The related ID fields act as foreign keys where shown. The Course entity connects enrollments, lessons, assignments, quizzes and payments.

8. CRUD Operations

CRUD stands for Create, Read, Update and Delete. These operations are used to manage records in the E-Learning database.

Operation	Example
Create	Add a new student, course, lesson or assignment.
Read	Display courses, enrolled students, lessons or results.
Update	Change course information, assignment status or student profile.
Delete	Remove obsolete records such as cancelled courses or unwanted notifications.

9. Joins and Subqueries

A JOIN combines information from two or more related tables using common key fields. For example, a JOIN can display student name, course title and enrollment date by combining Users, Enrollments and Courses.

Subqueries are queries written inside another query. They can be used to find students with high marks, courses having more enrollments than an average value, or assignments that have not yet been submitted.

- INNER JOIN – returns matching records from related tables.
- LEFT JOIN – returns all records from the left table and matching records from the right table.
- Subquery – useful for filtering or comparing values with calculated results.

10. Normalization – 3NF

Normalization organizes data into related tables and reduces unnecessary duplication. The E-Learning database separates students, courses, lessons, assignments, submissions and results into independent but connected tables.

The design aims for Third Normal Form (3NF), where non-key attributes depend on the key and unnecessary transitive dependencies are avoided.

- 1NF: Store atomic values and avoid repeating groups.
- 2NF: Remove partial dependency on part of a composite key.
- 3NF: Remove transitive dependency and keep attributes dependent on the primary key.
- Benefits: reduced redundancy, fewer update anomalies and improved consistency.

11. Advanced DBMS Features

Views, triggers and stored procedures can be used to provide reusable database functionality and automate selected operations.

View

A view such as `course_enrollment_view` can display course names together with the number of enrolled students.

Trigger

A trigger can automatically create a notification when a student successfully enrolls in a course. Flow: Enrollment Created → Trigger → Notification Record Created.

Stored Procedure

A stored procedure such as `EnrollStudent` can accept Student ID and Course ID, check required conditions, create the enrollment record and return the enrollment status.

12. Search and Reports

The system can provide search and reporting features using SQL queries over the relational database.

- Search student by name or email.
- Search course by title or instructor.
- View students enrolled in a course.
- View assignment submission status.
- View quiz results and marks.
- Course Enrollment Report.
- Student Performance Report.
- Assignment Submission Report.
- Payment Report.
- Course-wise Student Report.

13. Application Flow

The application can be presented as a simple learning management flow:

- Login / Registration
- Dashboard
- Browse Courses
- Course Enrollment
- View Lessons and Learning Materials
- Attempt Assignments / Quizzes
- Submit Answers
- View Results
- Payment / Enrollment Status
- Notifications and Reports

14. Conclusion

The E-Learning Management System demonstrates how a relational database can organize online education data in a structured manner. It connects students, instructors, courses, lessons, enrollments, assignments, submissions and assessments through primary and foreign keys.

The project provides practical implementation of DBMS concepts such as CRUD operations, joins, subqueries, normalization, views, triggers, stored procedures and reporting. It provides a suitable database-focused structure for a DBMS mini project.

15. Future Scope

- Mobile application support for students and instructors.
- Online video classes and live lecture integration.
- Automated certificate generation.
- Advanced analytics for student performance.
- AI-based course and learning recommendations.
- Online examination with automatic evaluation.
- Cloud deployment and scalable database infrastructure.
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